

DATA SCIENCE RESOURCES

Recommendation System Projects

Project 1. Book Recommendation System

Problem Statement: With thousands of books available online, users often struggle to find titles tailored to their interests or preferences.

Objective: Build a content-based and/or collaborative filtering book recommendation engine that suggests books based on user ratings or book metadata.

Dataset Source

- [Book-Crossing Dataset](#)
- [Goodreads user rating dataset](#)

Techniques & Concepts

- User-Item Matrix
- Cosine Similarity for content-based filtering
- KNN or Matrix Factorization (SVD, ALS)
- TF-IDF for metadata matching (genre, description, author)

Tools & Libraries: Python, Scikit-learn, Pandas, Surprise, Streamlit (optional)

Expected Outcome: An interactive system that allows users to input a book and receive top-N similar recommendations based on their taste or item profile.

Project 2. Product Suggestion Model (E-commerce)

Problem Statement: E-commerce platforms like Amazon need scalable product recommendation systems to personalize user experience and increase sales.

Objective: Build a product recommendation engine that suggests similar or complementary items based on user purchase history or item similarity.

Dataset Source

- [Amazon Product Dataset](#)
- [Retailrocket Recommender Dataset](#)

Techniques & Concepts

- Collaborative filtering (User-User and Item-Item)
- Market basket analysis (Apriori or FP-Growth)
- ALS matrix factorization (Spark MLlib)
- Content-based recommendations using metadata

Tools & Libraries: Python, Surprise, LightFM, PySpark, Scikit-learn

Expected Outcome: A model that suggests frequently bought together items, similar products, or cross-sell/up-sell combos based on user interaction data.

Project 3. Music Recommendation Engine

Problem Statement: Music platforms like Spotify and YouTube Music need intelligent systems to suggest songs based on listener behavior, mood, or audio features.

Objective: Create a personalized music recommender using song metadata, audio features, or listening patterns.

Dataset Source

- [Million Song Dataset](#)
- [Spotify Audio Features Dataset](#) (Kaggle)

Techniques & Concepts

- Content-based filtering using audio features (tempo, valence, energy)
- K-means clustering for genre/mood segmentation
- Matrix factorization for collaborative filtering
- Deep learning (autoencoders for embeddings)

Tools & Libraries: Python, Scikit-learn, Spotify (Spotify API), TensorFlow, Keras

Expected Outcome: A model that recommends similar tracks based on tempo, mood, genre, or listener similarity.

Project 4. News Article Recommender

Problem Statement: News websites want to improve user engagement by suggesting relevant articles based on past reading behavior and content interest.

Objective: Build a news recommender that serves relevant content using NLP techniques and user browsing history.

Dataset Source

- [MIND News Dataset](#) (Microsoft)
- [AG News Dataset](#) (Kaggle)

Techniques & Concepts

- NLP-based content similarity using TF-IDF or BERT embeddings
- Real-time clickstream-based collaborative filtering
- News topic clustering and personalization
- Multi-armed bandit (optional) for reinforcement

Tools & Libraries: Python, Scikit-learn, Hugging Face Transformers (for BERT), Faiss, FastAPI

Expected Outcome: A system that recommends news articles similar in topic or tone to a user's reading preferences or clicked articles.
